

Volcano多元算力管理Device Plugin

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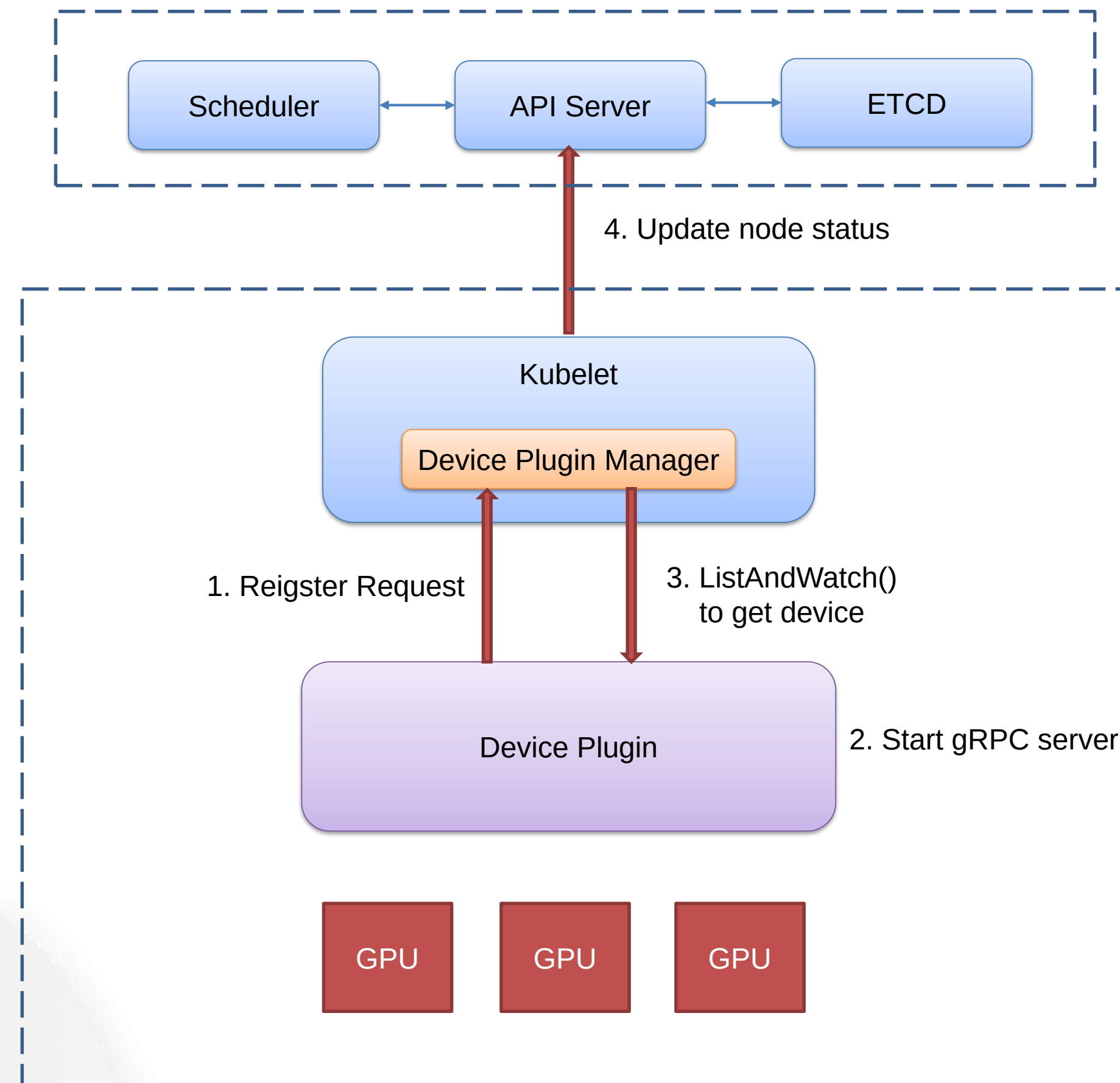


01

Kubernetes Device Plugin



Device Plugin介绍



Extended Resource: 通过这种方式，将资源的名称和总数量上报给API Server，Scheduler根据使用该资源的POD的创建、删除使用该资源，从而在调度时判断是否有满足资源条件的节点。

Device Plugin: Kubernetes提供device plugin 框架，用户可以使用该框架来上报特定的硬件资源到Kubelet。

Device Plugin 实现

```
service Registration {  
    rpc Register(RegisterRequest) returns (Empty) {}  
}
```

通过gRPC device plugin将注册到成kubelet中的service

```
service DevicePlugin {  
    rpc GetDevicePluginOptions(Empty) returns (DevicePluginOptions) {}  
    rpc ListAndWatch(Empty) returns (stream ListAndWatchResponse) {}  
    rpc Allocate(AllocateRequest) returns (AllocateResponse) {}  
    rpc PreStartContainer(PreStartContainerRequest) returns  
    (PreStartContainerResponse) {}  
}
```

Plugin, 利用在/var/lib/kubelet/device-plugins/路径下的Unix socket启动gRPC

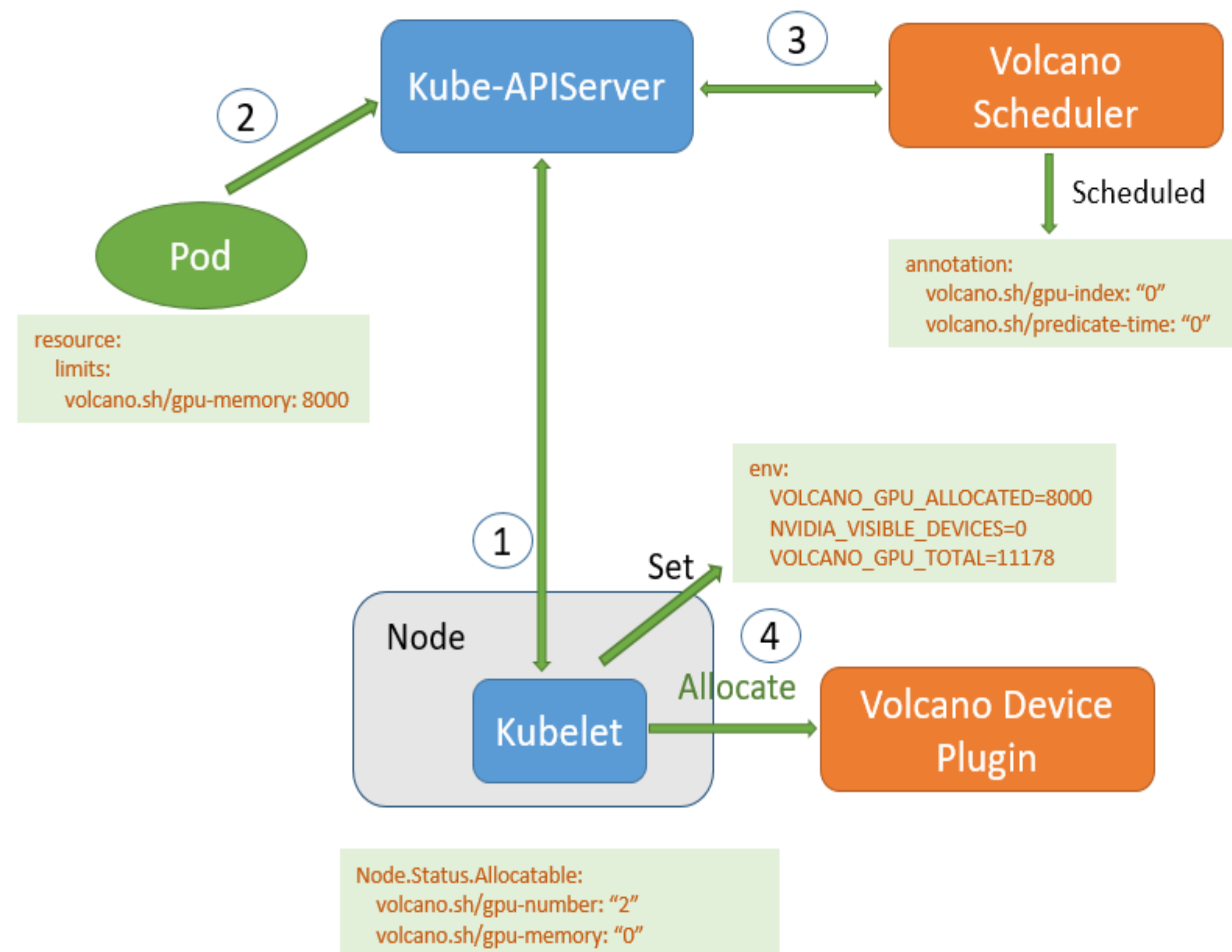


02

Volcano Device Plugin



GPU 共享



1. GPU device plugin collected and report GPU resources.
2. User submit a Pod which uses GPU resources by gpu-memory.
3. Volcano enable GPU predicate scheduler.
4. Start container and update pod annotations.

使用GPU共享

```
$ kubectl get node {node name} -oyaml
```

```
...
```

```
status:
```

```
addresses:
```

```
- address: 172.17.0.3
```

```
  type: InternalIP
```

```
- address: volcano-control-plane
```

```
  type: Hostname
```

```
allocatable:
```

```
  cpu: "4"
```

```
  ephemeral-storage: 123722704Ki
```

```
  hugepages-1Gi: "0"
```

```
  hugepages-2Mi: "0"
```

```
  memory: 8174332Ki
```

```
  pods: "110"
```

```
  volcano.sh/gpu-memory: "89424"
```

```
  volcano.sh/gpu-number: "8" # GPU resource
```

```
capacity:
```

```
  cpu: "4"
```

```
  ephemeral-storage: 123722704Ki
```

```
  hugepages-1Gi: "0"
```

```
  hugepages-2Mi: "0"
```

```
  memory: 8174332Ki
```

```
  pods: "110"
```

```
  volcano.sh/gpu-memory: "89424"
```

```
  volcano.sh/gpu-number: "8" # GPU resource
```

```
kubectl edit cm -n volcano-system volcano-scheduler-  
configmap
```

```
kind: ConfigMap
```

```
apiVersion: v1
```

```
metadata:
```

```
  name: volcano-scheduler-configmap
```

```
  namespace: volcano-system
```

```
data:
```

```
  volcano-scheduler.conf: |
```

```
    actions: "enqueue, allocate, backfill"
```

```
    tiers:
```

```
      - plugins:
```

```
        - name: priority
```

```
        - name: gang
```

```
        - name: conformance
```

```
      - plugins:
```

```
        - name: drf
```

```
        - name: predicates
```

```
          arguments:
```

```
            predicate.GPUSharingEnable: true # enable GPU
```

```
sharing
```

```
  - name: proportion
```

```
  - name: nodeorder
```

```
  - name: binpack
```



Join Volcano



Website: <https://volcano.sh/en/>



Github: <https://github.com/volcano-sh/volcano>



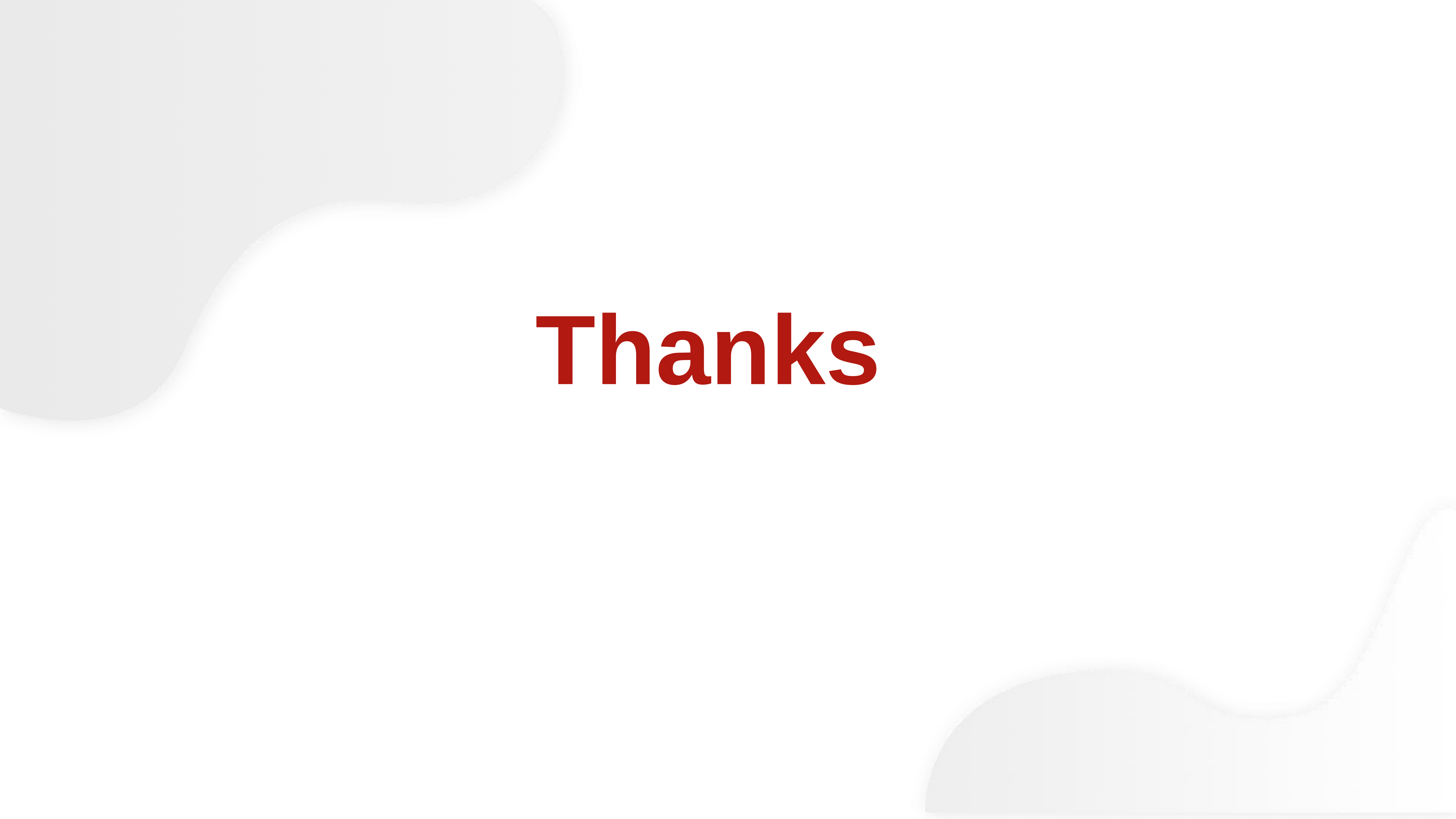
Slack Channel: <https://volcano-sh.slack.com/>



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Thanks